

What Was the Fed Thinking?

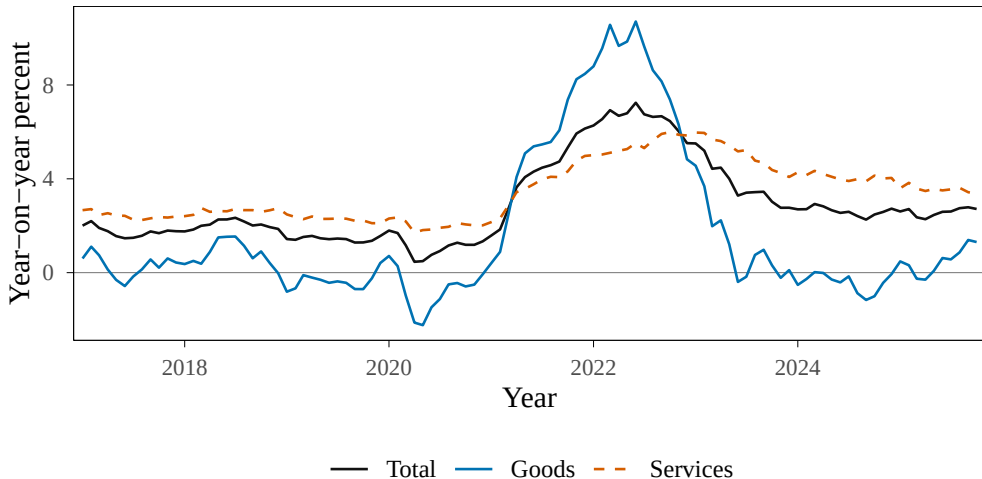
Optimal Monetary Policy under Capacity Constraints¹

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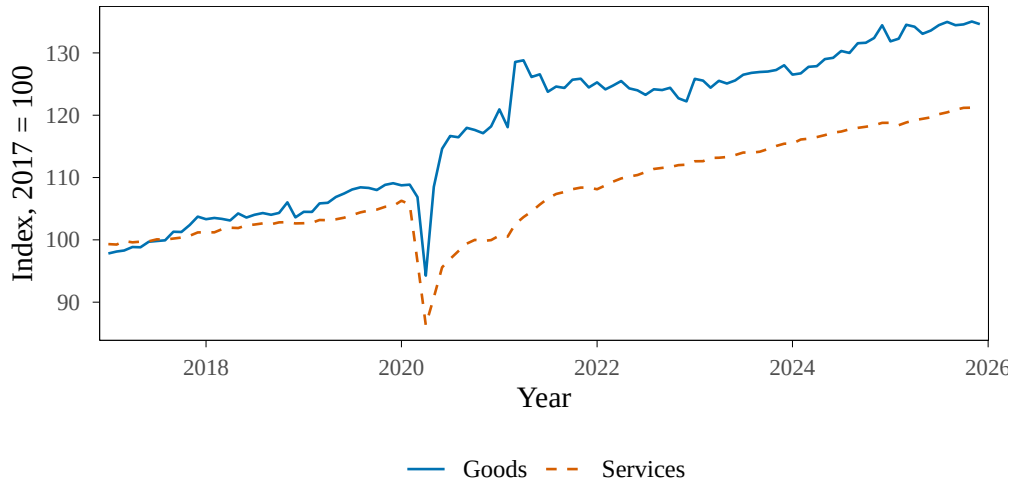
¹The views expressed are those of the authors and not necessarily those of the Federal Reserve Board or the Federal Reserve System.

US PCE Inflation



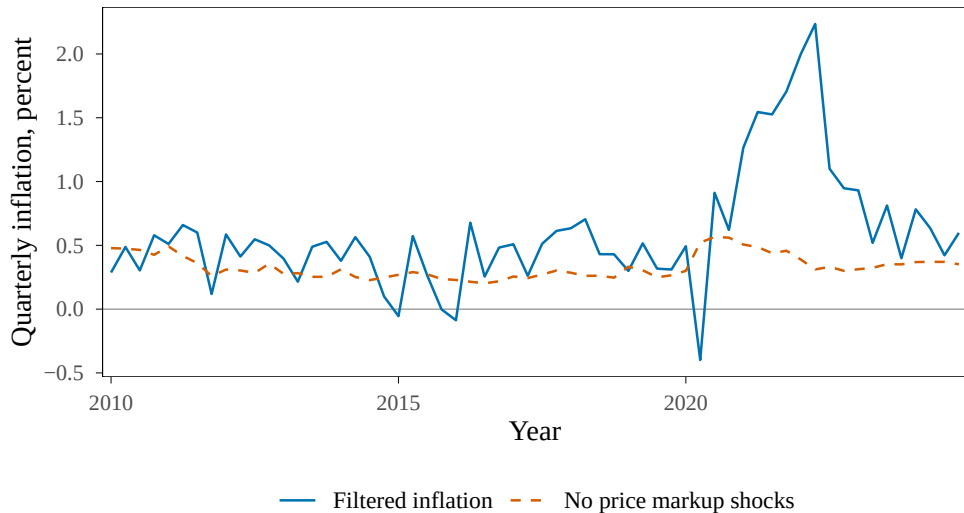
Source: Bureau of Economic Analysis and authors' calculations.

Real Consumer Expenditure



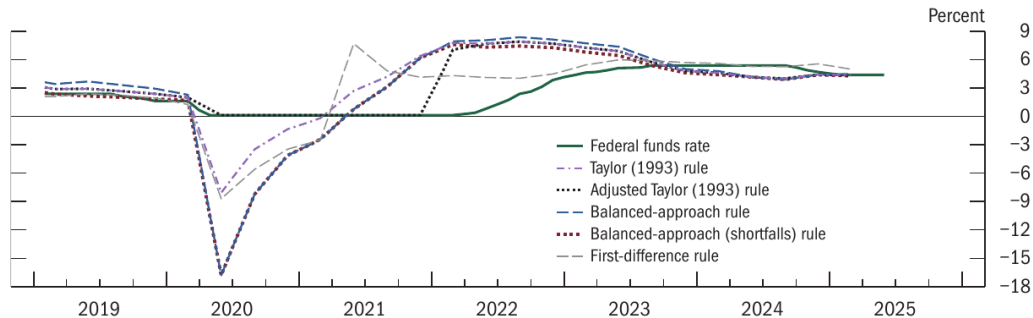
Source: Bureau of Economic Analysis and authors' calculations.

Role of Price Markup Shocks, Smets-Wouters Model



Fed Funds Rate versus Simple Policy Rules (from FRB MPR)

Figure A. Historical federal funds rate prescriptions from simple policy rules



Note: The rules use historical values of core personal consumption expenditures (PCE) inflation, the unemployment rate, and, where applicable, the midpoint of the target range for the federal funds rate constructed as the average of the lower and upper limits of the target range. Quarterly projections of longer-run values for the federal funds rate, the unemployment rate, and inflation used in the computation of the rules' prescriptions are interpolations to quarterly values of projections from the Survey of Market Expectations. The rules' prescriptions are quarterly, and the federal funds rate data are the monthly average of the daily midpoint of the target range for the federal funds rate.

What We Do

- ▶ Inflation surge during 2021-2022, driven by goods price inflation.
 - ▶ Demand was strong amid tightened supply chains.
- ▶ Fed response, per simple / estimated rules, was initially muted.
- ▶ In this paper we use an NK model with sector-level capacity constraints.
 - ▶ then allow uncertainty on the part of the Fed about the shocks.
- ▶ Question:
 - ▶ Do capacity constraints change the optimal monetary policy?
 - ▶ How did the Fed response compare to optimal policy?

Outline

1. Model
2. Methodology
3. Quantification
4. Central bank beliefs

Model

Overview

- ▶ Standard NK model.
- ▶ Two production sectors: goods and services.
- ▶ The wrinkle: binding capacity can cap goods output (sector 1):

$$Y_{1t} \leq \bar{Y}_{1t}.$$

- ▶ When binding, the capacity constraint shifts / steepens the Phillips curve.

Households

- ▶ The representative household:

$$\max \quad \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Theta_t \left[\frac{C_t^{1-\rho}}{1-\rho} - \frac{L_t^{1+\psi}}{1+\psi} \right].$$

$$\text{st} \quad P_t C_t + B_t = W_t L_t + (1 + i_{t-1}) B_{t-1} + \mathcal{D}_t.$$

- ▶ Aggregate consumption is a CES bundle of two sectoral goods.

$$C_t = \left[\bar{s}^{1/\vartheta} C_{1t}^{(\vartheta-1)/\vartheta} + (1 - \bar{s})^{1/\vartheta} C_{2t}^{(\vartheta-1)/\vartheta} \right]^{\vartheta/(\vartheta-1)}.$$

- ▶ Relative prices and demand.

$$R_{jt} \equiv P_{jt}/P_t, \quad 1 = \bar{s} R_{1t}^{1-\vartheta} + (1 - \bar{s}) R_{2t}^{1-\vartheta}, \quad C_{jt} = R_{jt}^{-\vartheta} C_t.$$

Firms

- ▶ Both sectors use labor-only technologies.

$$Y_{jt}(i) = Z_{jt}N_{jt}(i), \quad j \in \{1, 2\}.$$

- ▶ Sectoral marginal costs.

$$MC_{jt} = \frac{W_t/P_t}{Z_{jt}}.$$

- ▶ Sector 1 faces the hard output-capacity constraint.

$$Y_{1t} \leq \bar{Y}_{1t}.$$

Firms and Price Setting

- ▶ Each sector is monopolistically competitive, varieties aggregated with elasticity $\epsilon > 1$.
- ▶ Firms face Rotemberg price-adjustment costs.
- ▶ Firm problem ($\Pi_{jt} \equiv P_{jt}/P_{j,t-1}$):

$$\max_{\{P_{jt}(i)\}} \mathbb{E}_t \sum_{k=0}^{\infty} Q_{t,t+k} \left[\frac{P_{j,t+k}(i)}{P_{t+k}} Y_{j,t+k}(i) - MC_{j,t+k} Y_{j,t+k}(i) - \frac{\Phi}{2} (\Pi_{j,t+k}(i) - 1)^2 Y_{j,t+k} \right].$$

$$\text{st } Y_{jt}(i) = \left(\frac{P_{jt}(i)}{P_{jt}} \right)^{-\epsilon} Y_{jt}.$$

$$\text{and } Y_{1t}(i) \leq \bar{Y}_{1t}.$$

Sector 1 Capacity as a Phillips-Curve Wedge

- ▶ Let $\mu_{1t}(i) \geq 0$ be the multiplier on $Y_{1t}(i) \leq \bar{Y}_{1t}$.

$$\mu_{1t}(i) (\bar{Y}_{1t} - Y_{1t}(i)) = 0, \quad \mu_{1t}(i) \geq 0, \quad Y_{1t}(i) \leq \bar{Y}_{1t}.$$

- ▶ The multiplier enters the sector 1 pricing FOC like an extra marginal-cost term.

$$\Phi(\Pi_{1t} - 1)\Pi_{1t} = (\epsilon - 1)(MC_{1t}/R_{1t} - 1) + \epsilon\mu_{1t} + \beta\mathbb{E}_t \frac{Q_{t,t+1}Y_{1,t+1}}{Q_{t,t}Y_{1t}} \Phi(\Pi_{1,t+1} - 1)\Pi_{1,t+1}.$$

- ▶ After log-linearization, binding capacity looks like a markup shock.

$$\pi_{1t} = \beta\mathbb{E}_t\pi_{1,t+1} + \kappa \left(\widehat{mc}_{1t} + \widehat{\mathcal{M}}_{1t} - \widehat{r}_{1t} \right) + \chi \widehat{\mu}_{1t}, \quad \kappa = \frac{\epsilon - 1}{\Phi}, \quad \chi = \frac{\epsilon}{\Phi}.$$

$$\text{Slack: } \widehat{\mu}_{1t} = 0. \quad \text{Binding: } \widehat{\mu}_{1t} > 0.$$

Nonlinear Pricing

- ▶ If slack, Phillips curve is the standard one.

$$\Phi(\Pi_{jt} - 1)\Pi_{jt} = (\epsilon - 1)(MC_{jt}/R_{jt} - 1) + \beta \mathbb{E}_t \frac{Q_{t,t+1} Y_{j,t+1}}{Q_{t,t} Y_{jt}} \Phi(\Pi_{j,t+1} - 1)\Pi_{j,t+1}.$$

- ▶ Binding capacity replaces sector-1 pricing equation with $Y_{1t} = \bar{Y}_{1t}$.
- ▶ There is an additional pricing residual when the sector 1 constraint binds.

Slack sector 1: pricing residual = 0.

Binding sector 1: $Y_{1t} = \bar{Y}_{1t}$.

When $Y_{1t} = \bar{Y}_{1t}$, the Aggregate Phillips Curve also Steepens

- ▶ Aggregate inflation is approximately expenditure-weighted sectoral inflation.
- ▶ When $Y_{1t} = \bar{Y}_{1t}$, sector-1 demand pins down relative prices:

$$rp_{1t} = \frac{1}{\vartheta}(y_t - \bar{y}_{1t}), \quad -rp_{2t} = \frac{\bar{s}}{\vartheta(1 - \bar{s})}(y_t - \bar{y}_{1t}).$$

- ▶ Sector 2 is unconstrained, but its marginal cost inherits the relative-price pressure.
- ▶ This implies a steeper aggregate Phillips curve:

$$\pi_t = \pi_t^n + \Gamma_C(y_t - y_t^n) + \beta \mathbb{E}_t \pi_{2,t+1},$$

$$\Gamma_C = \frac{\epsilon - 1}{\Phi} \left(\rho + \psi + \frac{\bar{s}}{\vartheta(1 - \bar{s})} \right) + \frac{\bar{s}}{\vartheta(1 - \bar{s})} > \Gamma_U = \frac{(\epsilon - 1)(\rho + \psi)}{\Phi}.$$

Shocks

- Demand, productivity (sector), markups (sector), policy, and capacity.

$$\log \Theta_t = \rho_\theta \log \Theta_{t-1} + \frac{\sigma_\theta}{100} \varepsilon_t^\theta.$$

$$\log Z_{jt} = \rho_{zj} \log Z_{j,t-1} + \frac{\sigma_{zj}}{100} \varepsilon_t^{zj},$$

$$\log \mathcal{M}_{jt} = \rho_{\mu j} \log \mathcal{M}_{j,t-1} - \frac{\sigma_{\mu j}}{100} \varepsilon_t^{\mu j}.$$

- Capacity shocks are persistent innovations to $\bar{Y}_{1t}$.
- In the IRF calibration, steady-state excess capacity $\bar{Y}_1^{ss}$ is 5 percent.

$$\log \bar{Y}_{1t} = (1 - \rho_{\bar{Y}}) \log \bar{Y}_1^{ss} + \rho_{\bar{Y}} \log \bar{Y}_{1,t-1} + \frac{\sigma_{\bar{Y}}}{100} \varepsilon_t^{\bar{Y}}.$$

Policy Closure

- ▶ Taylor rule:

$$i_t = \alpha_i i_{t-1} + \alpha_\pi \pi_t + \alpha_y y_t + e_t^i.$$

- ▶ IRFs use $(\alpha_i, \alpha_\pi, \alpha_y) = (0, 1.5, 0.125)$.
 - ▶ Estimate coefficients in quantification.
- ▶ Optimal policy (next).

Solution Approach

Reduced Log-Linear System

- ▶ Log-linearize around steady state.
- ▶ Slack regime keeps the sector-1 pricing equation, binding regime imposes $y_{1,t} = \bar{y}_{1,t}$.
- ▶ This delivers a time-varying linear system used below.

$$A_t y_t = C_t + B_t y_{t-1} + D_t \mathbb{E}_t y_{t+1} + F_t \varepsilon_t.$$

- ▶ The binding path is solved jointly with policy (Guerrieri/Iacoviello OccBin).

$$b_t = \begin{cases} 0, & y_{1,t}^{sh} < \bar{y}_{1,t}, \\ 1, & y_{1,t}^{sh} \geq \bar{y}_{1,t}. \end{cases}$$

- ▶ With unconstrained terminal condition and path $\{b_t\}_{t \geq 0}$, can derive time-varying VAR.

$$y_t = J_t + Q_t y_{t-1} + G_t \varepsilon_t$$

Towards Optimal Policy: Define the Private-Sector Block

- ▶ Ongoing work on anticipated structural changes by/with Mariano Kulish

- ▶ From

$$A_t y_t = C_t + B_t y_{t-1} + D_t \mathbb{E}_t y_{t+1} + F_t \varepsilon_t,$$

- ▶ Remove the policy-rule row. This gives the *private-sector* system

$$A_t^{\text{ps}} y_t = C_t^{\text{ps}} + B_t^{\text{ps}} y_{t-1} + D_t^{\text{ps}} \mathbb{E}_t y_{t+1} + F_t^{\text{ps}} \varepsilon_t.$$

Central Bank Problem

- ▶ Given candidate capacity-regime path $\{b_t\}_{t \geq 0}$, central bank chooses allocation and i_t
- ▶ to penalize π_t gaps and efficient-output gaps, and smooth i_t :

$$\min_{\{y_t\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{1}{2} \pi_t^2 + \frac{0.25}{2} (y_t - y_t^e)^2 + \frac{w_{\Delta i}}{2} (i_t - i_{t-1})^2 \right].$$

- ▶ Benchmark IRFs set $w_{\Delta i} = 0$.
- ▶ subject to the private-sector equilibrium imposed as a constraint:

$$A_t^{\text{ps}} y_t = C_t^{\text{ps}} + B_t^{\text{ps}} y_{t-1} + D_t^{\text{ps}} \mathbb{E}_t y_{t+1} + F_t^{\text{ps}} \varepsilon_t.$$

FOC

- ▶ Use W_t to write the period loss in quadratic form:

$$\frac{1}{2}y_t'W_ty_t$$

- ▶ Let μ_t denote multipliers on private-sector equilibrium.
- ▶ And so the planner's FOC for y_t is

$$W_ty_t + (A_t^{\text{ps}})'\mu_t - \beta \mathbb{E}_t[(B_{t+1}^{\text{ps}})'\mu_{t+1}] - \beta^{-1}(D_{t-1}^{\text{ps}})'\mu_{t-1} = 0.$$

Augmented State, Backward Recursion and Optimal Policy Reduced-Form

- ▶ Stack allocations and multipliers:

$$\xi_t = \begin{pmatrix} y_t \\ \mu_t \end{pmatrix}.$$

- ▶ Private-sector equations plus planner FOCs form another time-varying linear system:

$$\tilde{A}_t \xi_t = \tilde{C}_t + \tilde{B}_t \xi_{t-1} + \tilde{D}_t \mathbb{E}_t \xi_{t+1} + \tilde{F}_t \varepsilon_t.$$

- ▶ Can do backward recursion on this to get the augmented reduced form:

$$\xi_t = \tilde{J}_t + \tilde{Q}_t \xi_{t-1} + \tilde{G}_t \varepsilon_t.$$

- ▶ Use an OccBin-style fixed point over the capacity-regime path b_t .

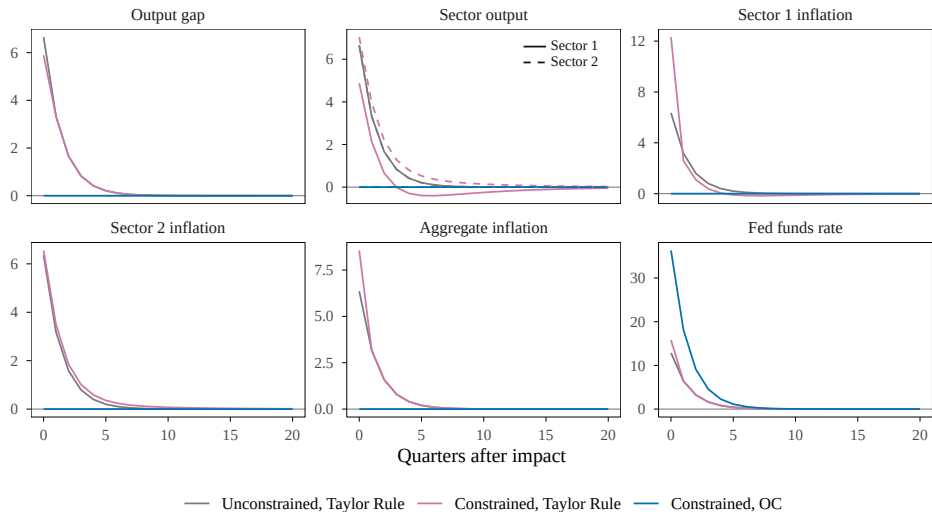
Estimation

Initial Estimation Strategy

- ▶ Estimate with the Taylor rule (and no central bank shock uncertainty).
- ▶ Observables include fed funds rate, output gap, sector output growth, sector inflation.
- ▶ Estimate shocks, steady-state capacity, and Taylor rule coefficients.
- ▶ Also jointly estimate discrete capacity durations.
- ▶ Counterfactual: removing capacity shocks explains $\sim$ half of goods inflation over covid
 - ▶ consistent with other work.

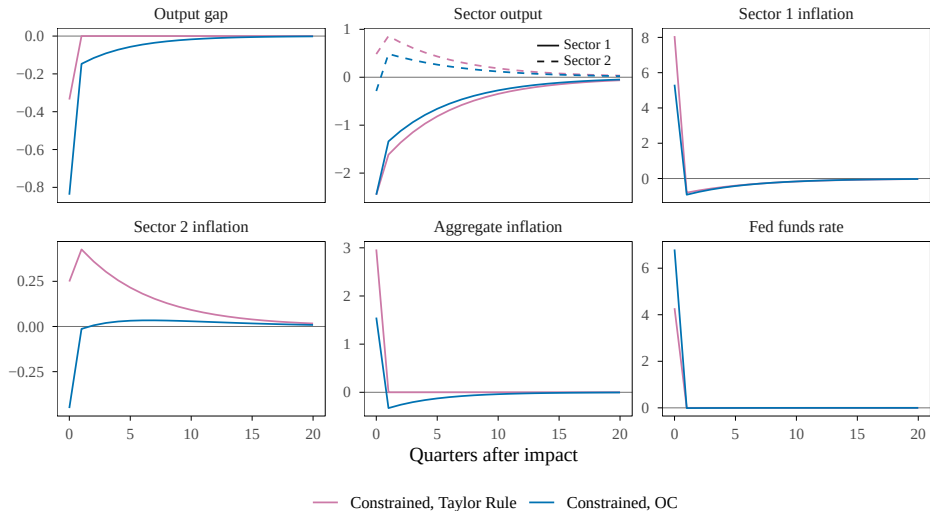
Impulse Responses

Discount Factor Shock



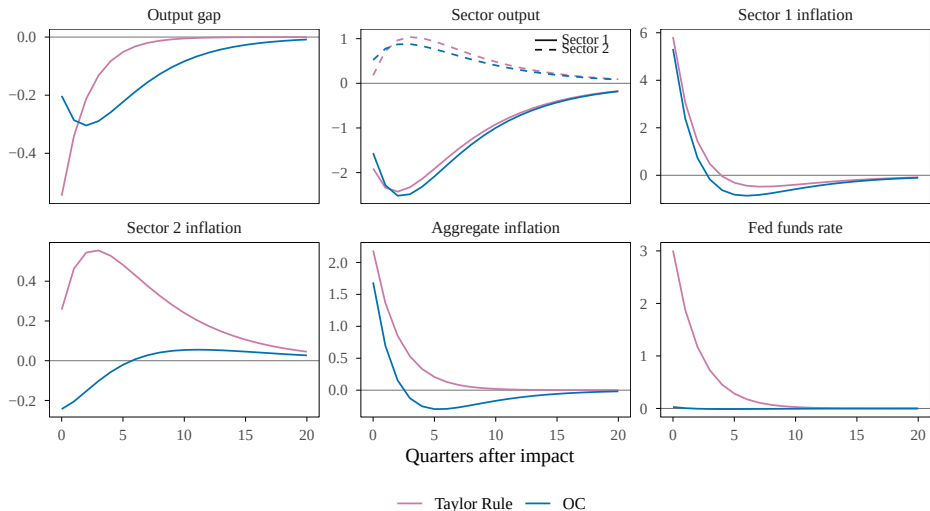
Demand shock: OC can use the nominal rate to undo the shock

Capacity Shock



Binding capacity shock: policy cannot restore sector-1 output but can create slack

Sector-1 Markup Shock



Fully offsetting inflation would require inefficient slack, so OC responds modestly

Interpreting the IRFs

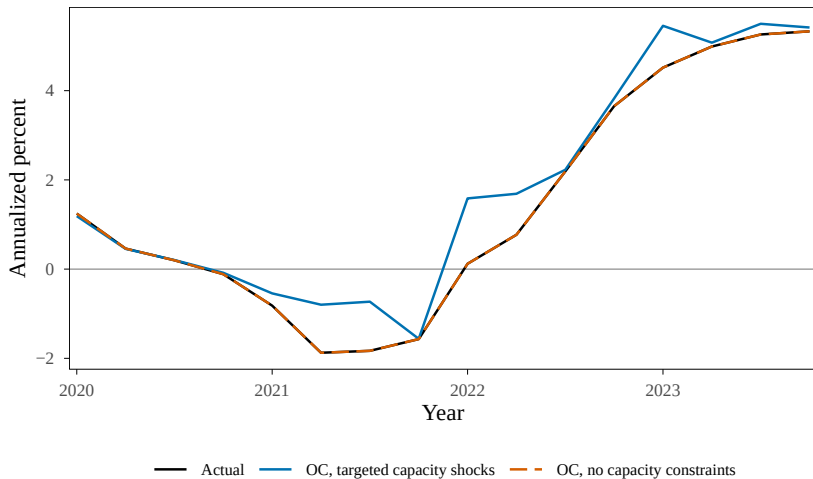
- ▶ Demand shock: OC can use the nominal rate to undo the shock.
 - ▶ because the benchmark loss has no direct interest-rate penalty.
- ▶ Binding capacity shock: policy cannot restore sector-1 output.
 - ▶ can create aggregate slack to offset relative-price inflation pressure.
- ▶ Sector-1 markup shock:
 - ▶ fully offsetting inflation would require inefficient slack, so OC responds modestly.
- ▶ Limited-information implication:
 - ▶ a capacity shock may be seen as a markup shock, muting the response.

Central Bank Uncertainty

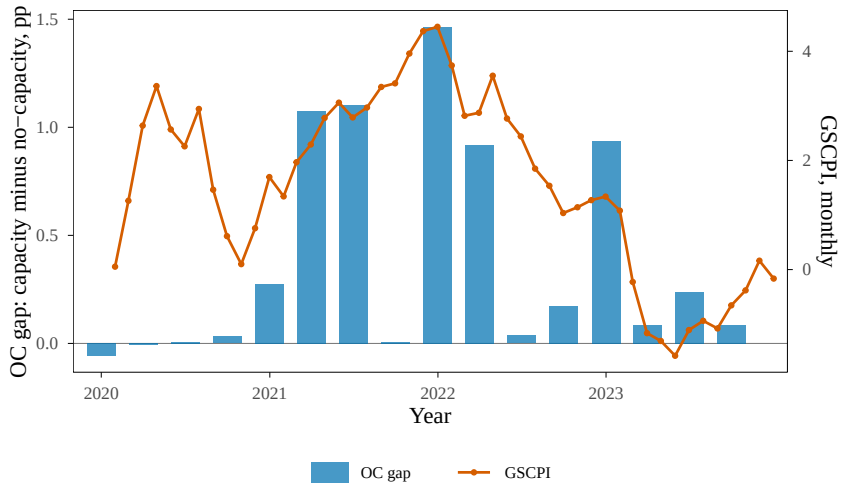
Comparing Fed Funds Rate to Optimal Policy

- ▶ Illustrative exercise.
- ▶ Filter data for two sets of shocks:
 1. Set of shocks assuming no capacity constraints.
 2. Set of shocks with capacity constraints.
- ▶ Choose loss weight $w_{\Delta i}$ each t so the no-capacity OC path matches the actual rate.
- ▶ Hold those weights fixed and recompute OC under estimated shocks with capacity.
- ▶ The gap between the OC paths isolates the effect of the shock interpretation.
- ▶ If capacity is binding, the same preferences imply a higher optimal rate.

Policy Rates with Time-Varying Weights



OC Gap and FRBNY Global Supply Chain Pressure Index



Central Bank Uncertainty about Shocks

- ▶ The central bank observes noisy signals about the latent structural shocks:

$$m_t = S\varepsilon_t + \nu_t.$$

- ▶ The signals imply posterior distribution over structural shocks.
- ▶ This posterior distribution maps into weights $\alpha_{t,d}$ where d indexes capacity duration.
- ▶ The policy rate is common across d scenarios and is chosen, given $\alpha_{t,d}$, to:

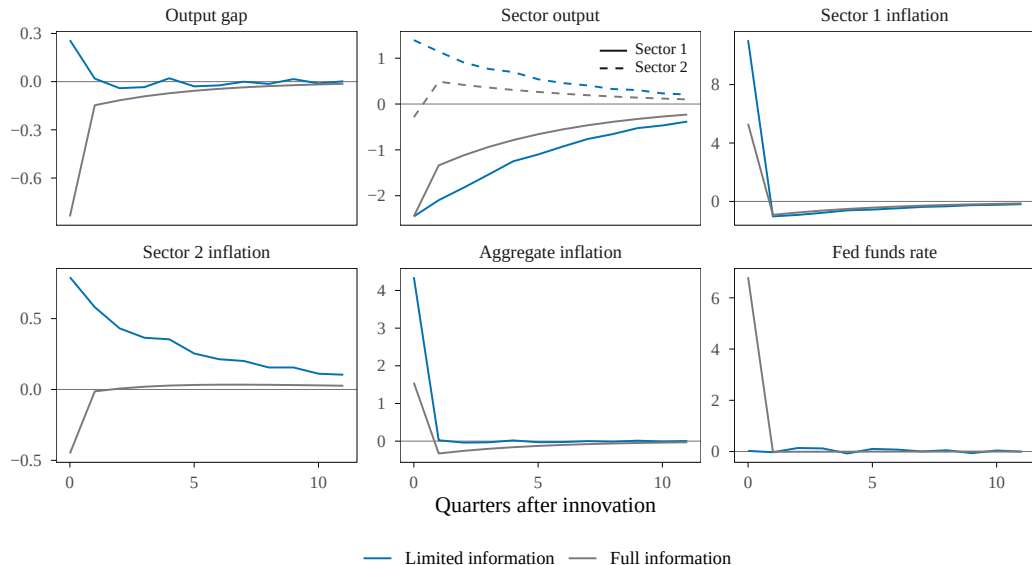
$$\min_{\{i_j\}, \{y_{d,j}, \pi_{d,j}, \dots\}_{d,j}} \sum_d \alpha_{t,d} \sum_{j=0}^{\infty} \beta^j \frac{1}{2} [\pi_{d,j}^2 + 0.25(y_{d,j} - y_{d,j}^e)^2].$$

- ▶ subject to each scenario's private-sector equilibrium.
- ▶ To use the methodology above, stack allocation for each possible duration d
 - ▶ with a single common interest rate across all scenarios.

Beliefs Experiment IRF

- ▶ The true shock is an adverse capacity innovation.
- ▶ The central bank receives a markup-like signal.
- ▶ It assigns no impact probability to capacity shocks.
- ▶ Actual economy binds for one quarter.

Beliefs Experiment IRF

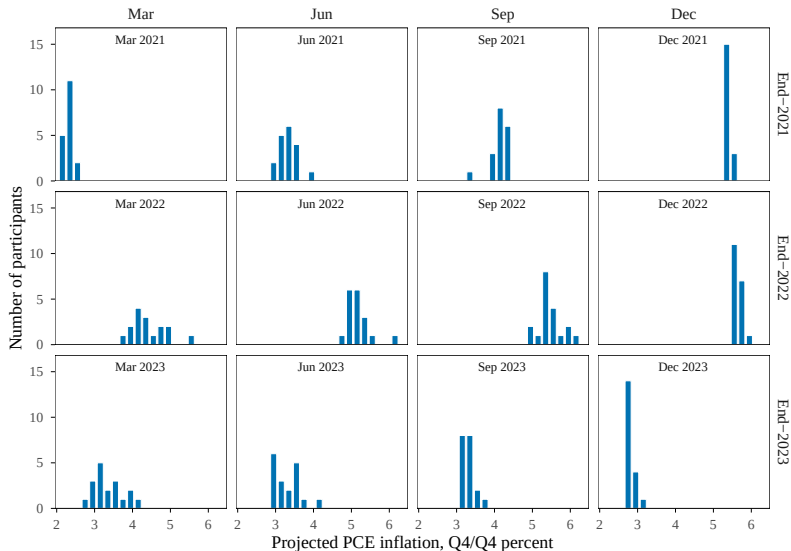


SEP Data

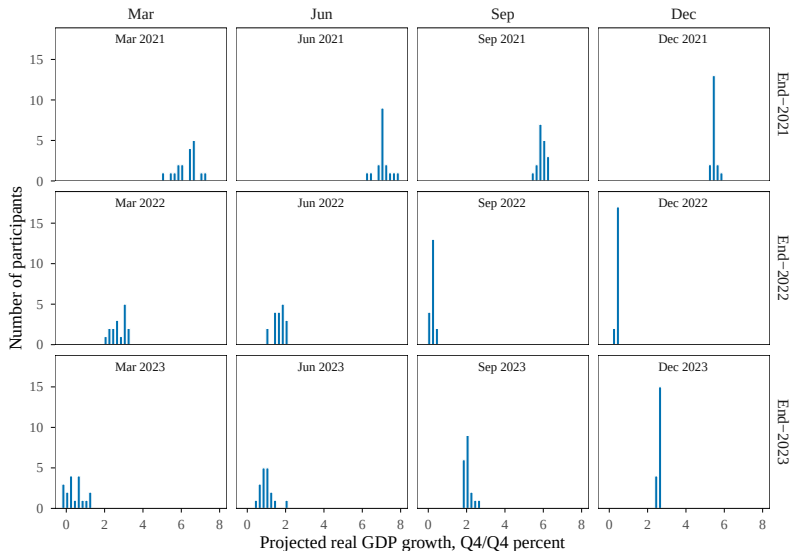
Statement of Economic Projections

- ▶ SEP: a real-time distribution of FOMC participant projections.
- ▶ Relevant objects: end-year PCE inflation, real GDP growth, and the fed funds rate.
- ▶ SEP projections are possibly informative moments of policymakers' beliefs.
- ▶ Early-year projections are especially useful:
 - ▶ inflation and output projections summarize perceived economic conditions,
 - ▶ rate projections summarize the policy response thought appropriate under those beliefs.

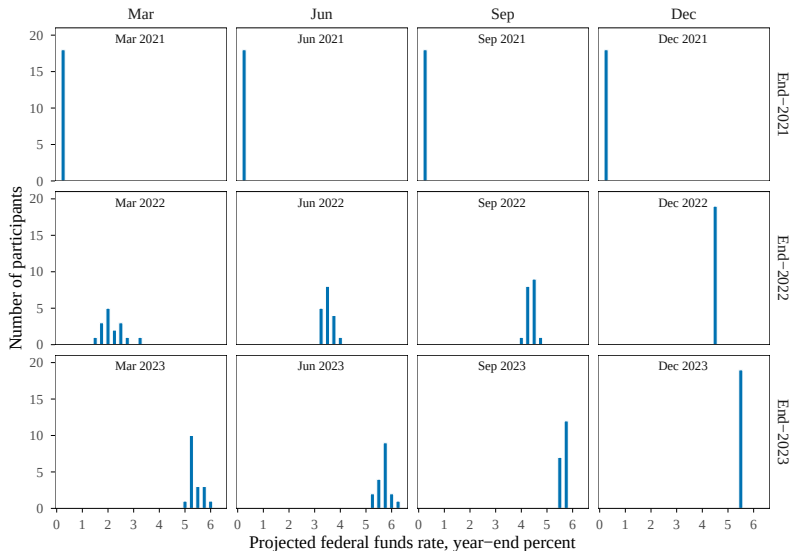
SEP Inflation Projections



SEP Real GDP Growth Projections



SEP Fed Funds Projections



Using the SEP to Estimate Signals

- ▶ Let s_t^{SEP} collect SEP moments of π_t , growth, and i_t projections.

$$s_t^{SEP} = H^{SEP} \mathbb{E}_t^{CB}(x_{t+h}) + \varepsilon_t^{SEP}.$$

- ▶ The same structural state can imply different SEP moments depending on central-bank beliefs:

$$\mathbb{E}_t^{CB}(x_{t+h}) = \sum_d \alpha_{t,d} \mathbb{E}_t(x_{d,t+h} \mid d).$$

- ▶ Adding SEP moments to the likelihood may be informative the signal process:

$$\ell(\Theta) = \ell_{\text{macro}}(\Theta) + \ell_{\text{SEP}}(\Theta).$$

Conclusions and Next Steps

- ▶ Model with capacity constraints has very different optimal policy implications.
- ▶ Estimate a central-bank signal process using macro observables and SEP moments.
- ▶ Use the estimated beliefs to solve the common-rate commitment problem.
- ▶ Compare actual to optimal policy under full information and SEP-disciplined beliefs.

Appendix

What is π_t^n When Capacity Binds?

- ▶ When sector 1 binds, its Phillips curve is replaced by the capacity equation:

$$Y_{1t} = \bar{Y}_{1t}.$$

- ▶ Sector-1 demand then pins down the relative price:

$$rp_{1t} = \frac{1}{\vartheta}(y_t - \bar{y}_{1t}).$$

- ▶ Aggregate inflation inherits this required relative-price movement:

$$\pi_t = \pi_{2t} + \frac{\bar{s}}{1 - \bar{s}}(rp_{1t} - rp_{1,t-1}).$$

- ▶ π_t^n is the natural relative-price inflation required by this scarcity.

$$\pi_t^n = \frac{\bar{s}}{1 - \bar{s}} \left[\frac{1}{\vartheta}(y_t^n - \bar{y}_{1t}) - rp_{1,t-1} \right].$$

Where Does π_t^n Enter?

- ▶ π_t^n is an aggregate Phillips-curve shifter induced by the constrained sector-1 relative price.

- ▶ Relaxed regime:

$$\pi_t = \Gamma_U(y_t - y_t^n) + \beta \mathbb{E}_t \pi_{t+1}.$$

- ▶ Binding regime:

$$\pi_t = \pi_t^n + \Gamma_C(y_t - y_t^n) + \beta \mathbb{E}_t \pi_{2,t+1}.$$

- ▶ So π_t^n enters the aggregate Phillips curve directly only when capacity binds.

Does Sector 2's Phillips Curve Steepen?

- ▶ The sector-2 pricing equation itself is unchanged:

$$\pi_{2t} = \beta \mathbb{E}_t \pi_{2,t+1} + \kappa (mc_{2t} - rp_{2t}).$$

- ▶ When sector 1 binds, sector-1 demand pins down its relative price:

$$rp_{1t} = \frac{1}{\vartheta} (y_t - \bar{y}_{1t}).$$

- ▶ The price-index restriction then pins down sector 2's relative price:

$$rp_{2t} = -\frac{\bar{s}}{1 - \bar{s}} rp_{1t} = -\frac{\bar{s}}{\vartheta(1 - \bar{s})} (y_t - \bar{y}_{1t}).$$

- ▶ Hence sector 2's real marginal cost includes an extra output-sensitive term:

$$mc_{2t} - rp_{2t} = mc_{2t} + \frac{\bar{s}}{\vartheta(1 - \bar{s})} (y_t - \bar{y}_{1t}).$$

- ▶ The sector-2 FOC is unchanged; the aggregate reduced-form Phillips curve is steeper.

Solving the Actual Path

- ▶ The common-rate problem delivers a planned policy path $\{i_j^{CB}\}_{j \geq 0}$.
- ▶ The actual path is solved separately: it is not a posterior-weighted average.
- ▶ Take the true shock path and inherited state:

$$\{\varepsilon_{t+j}^{true}\}_{j \geq 0}, \quad x_{t-1}.$$

- ▶ Guess a capacity-regime path $\{b_j^{act}\}_{j \geq 0}$.
- ▶ Given $\{b_j^{act}\}$, solve the time-varying private-sector LRE system with $\{i_j^{CB}\}$ imposed.
- ▶ Update the binding path from shadow sector-1 output:

$$b_j^{new} = \mathbf{1}\{y_{1,t+j}^{shadow} \geq \bar{y}_{1,t+j}\}.$$

- ▶ Iterate until $b_j^{new} = b_j^{act}$, then update beliefs next period.

Eggertsson-Woodford Check

- ▶ The EW result is the canonical history-dependent optimal-policy result.
- ▶ A negative natural-rate shock makes the ZLB bind.
- ▶ Optimal commitment promises future accommodation after the shock ends.
- ▶ That promise raises expected inflation and reduces the initial slump.
- ▶ Clean diagnostic for time-varying OC machinery.

EW Benchmark

- ▶ Private-sector block:

$$x_t = \mathbb{E}_t x_{t+1} - \sigma (i_t - \mathbb{E}_t \pi_{t+1} - r_t^n), \quad \pi_t = \kappa x_t + \beta \mathbb{E}_t \pi_{t+1}.$$

- ▶ The policy rate is a deviation from steady state.

$$i_t^{level} = i_t + r^*, \quad i_t \geq -r^*.$$

- ▶ Planner objective:

$$\min_{\{x_t, \pi_t, i_t\}} \frac{1}{2} \sum_{t=0}^{T-1} \beta^t (\pi_t^2 + \lambda_x x_t^2).$$

Applying the TVP Method

- ▶ Guess a ZLB duration L .
- ▶ For $t < L$, impose the binding-regime equation $i_t = -r^*$.
- ▶ For $t \geq L$, leave i_t as the planner's instrument.
- ▶ Conditional on L , the system is linear time-varying.

$$\begin{bmatrix} Q & A'_L \\ A_L & 0 \end{bmatrix} \begin{pmatrix} z \\ \mu \end{pmatrix} = \begin{pmatrix} 0 \\ b_L \end{pmatrix}.$$

- ▶ Enumerate L , keep feasible paths, choose the one with lowest loss.

Simulation

- ▶ Natural-rate shock lasts four quarters.
- ▶ Discretion keeps rates at zero for four quarters.
- ▶ Optimal commitment keeps rates at zero for five quarters.
- ▶ The extra zero-rate quarter is chosen by the OC problem, not imposed.
- ▶ Initial output gap: -2.17 percent under OC versus -4.20 under discretion.

